

Erdős Problem #978

EP-978	L1 - Tractable	Partiellement résolu	Number Theory
--------	----------------	----------------------	---------------

Énoncé original (anglais, reproduction mot pour mot)

Let $f \in \mathbb{Z}[x]$ be an irreducible polynomial of degree $k > 2$ (and suppose that $k \equiv 2^l$ for any $l \geq 1$) such that the leading coefficient of f is positive.

Does the set of integers $n \geq 1$ for which $f(n)$ is $(k-1)$ -power-free have positive density?

Are there infinitely many n for which $f(n)$ is $(k-2)$ -power-free?

In particular, does $n^4 + 2$ represent infinitely many squarefree numbers?

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>